sPHENIX Internal Note

Pileup Correction to Vertex + MBD Acceptance (Luminosity Note Section 3)

Isolated Photon Cross-Section in p+p at $\sqrt{s} = 200$ GeV

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Abstract

This note quantifies the pileup correction to the vertex + MBD coincidence acceptance in the sPHENIX isolated photon cross-section measurement (PPG12). In double-interaction events, the overlay collision provides additional MBD hits and shifts the reconstructed vertex toward the beam center, increasing the effective acceptance from $\varepsilon_{\mathrm{vtx+MBD}} = 0.68$ (single interaction) to 0.81 (double interaction) — a 19% relative increase. After Poisson weighting with the measured double-interaction fractions, the net correction to the cross-section is -2.1% for 0 mrad ($f_{\mathrm{double}} = 11.1\%$) and -0.7% for 1.5 mrad ($f_{\mathrm{double}} = 3.9\%$), with a mild p_{T}-dependence rising from $+2\%$ at 8 GeV to $+3.5\%$ at 30 GeV. This correction is currently not applied in the production efficiency chain, which uses single-interaction MC only.

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1 Motivation

The sPHENIX luminosity note (Section 3) describes a correction needed when the analysis requires both an MBD coincidence and a z-vertex selection. In PPG12, events must satisfy:

- MBD N&S ≥ 1 (at least one hit on each side)
- $|z_{\rm vtx}^{\rm reco}| < 60$ cm

The production efficiency chain (`RecoEffCalculator_TTreeReader.C` → `CalculatePhotonYield.C`) computes $\varepsilon_{\rm vtx+MBD}$ from single-interaction MC:

$$\varepsilon_{\rm vtx+MBD} = \frac{N(\text{truth photons passing MBD + vertex cut})}{N(\text{truth photons with } |z_{\rm vtx}^{\rm truth}| < 60 \text{ cm})} \tag{1}$$

In data, pileup events modify this acceptance in two ways:

1. **MBD rescue:** If the signal collision fires only one MBD side (or neither), the overlay collision can provide the missing hits, satisfying the coincidence requirement.
2. **Vertex centering:** The MBD-reconstructed vertex is approximately the average of two collision vertices. This pulls the vertex toward zero, increasing the fraction passing the $|z_{\rm vtx}| < 60$ cm cut.

The correction is:

$$\varepsilon_{\rm vtx+MBD}^{\rm blended} = (1 - f_{\rm double}) \times {\rm Pass}_1 + f_{\rm double} \times {\rm Pass}_2 \tag{2}$$

where Pass_1 is the single-interaction acceptance and Pass_2 is the double-interaction acceptance, and $f_{\rm double}$ is the event-level double-interaction fraction from Poisson statistics.

2 Method

2.1 Samples

Single-interaction MC uses the three standard photon-jet samples, weighted by production cross-section:

- `photon5` (0–14 GeV), weight $= \sigma_5/\sigma_{20} = 1122$
- `photon10` (14–30 GeV), weight $= \sigma_{10}/\sigma_{20} = 53.2$
- `photon20` (30+ GeV), weight $= 1.0$ (reference)

Double-interaction MC uses the full GEANT4 double-interaction sample:

- `photon10_double` — two independent pp collisions simulated per event, with the photon from the primary collision in the 10–100 GeV truth $p_{\rm T}$ range.

Two million events are processed per sample.

2.2 Truth photon selection

Standard PPG12 truth criteria: `pid` $= 22$, `photonclass` < 3, $E_{\rm T}^{\rm iso,truth} < 4$ GeV (cone $R = 0.3$), $|\eta| < 0.7$, $8 \leq p_{\rm T} < 36$ GeV.

The denominator of Eq. (1) requires $|z_{\rm vtx}^{\rm truth}| < 60$ cm (same as `RecoEffCalculator_TTreeReader.C:1694`).

2.3 Acceptance categories

For each truth photon in the denominator, we check four acceptance conditions independently:

1. **MBD + vertex (combined):** MBD N&S ≥ 1 AND $|z_{\rm vtx}^{\rm reco}| < 60$ cm — this is $\varepsilon_{\rm vtx+MBD}$ used in the cross-section formula.
2. **MBD only:** MBD N&S ≥ 1 (no reco vertex cut).
3. **Vertex only:** $|z_{\rm vtx}^{\rm reco}| < 60$ cm (no MBD requirement).
4. **MBD hit decomposition:** classified into both N&S, N only, S only, neither.

2.4 Pileup fractions

Event-level double-interaction fractions from `calc_pileup_range.C` (Poisson statistics, lumi-weighted):

- 0 mrad: $f_{\rm double} = 11.1\%$ (event-level; 22.4% cluster-weighted)
- 1.5 mrad: $f_{\rm double} = 3.9\%$ (event-level; 7.9% cluster-weighted)

Triple+ interactions are folded into the double fraction ($\text{Prob}_3 < 0.3\%$ for $\mu < 0.3$).

3 Results

3.1 Single vs. double-interaction acceptance

Figure 1 shows the per-$p_{\rm T}$ vertex+MBD acceptance for single-interaction (blue) and double-interaction (red) MC. The double-interaction acceptance is systematically higher across all $p_{\rm T}$ bins, with a ratio of $\sim$1.19 (Figure 2).

3.2 Acceptance breakdown

Figure 3 separates the MBD and vertex contributions:

- **MBD acceptance:** increases from 69% (single) to 92% (double) — a 33% relative increase. The overlay collision rescues events where the signal fired only one MBD side or neither.
- **Vertex acceptance:** increases from 75% (single) to 83% (double) — an 11% relative increase. The averaged vertex is narrower than individual vertices.

The MBD rescue effect dominates the correction.

3.3 MBD hit decomposition

Figure 4 shows the stacked MBD hit fractions. In single-interaction MC, $\sim$8% of truth photons produce no MBD hits ("neither"), consistent with the GEANT4 prediction ($P_{\rm neither} = 8.8\%$). In double-interaction MC, this drops to $< 1\%$ because the overlay almost always provides some MBD activity.

3.4 Correction factors

Figure 5 shows the per-$p_{\rm T}$ correction factor $\varepsilon_{\rm vtx+MBD}^{\rm blended}/\varepsilon_{\rm vtx+MBD}^{\rm single}$ for both crossing-angle periods.

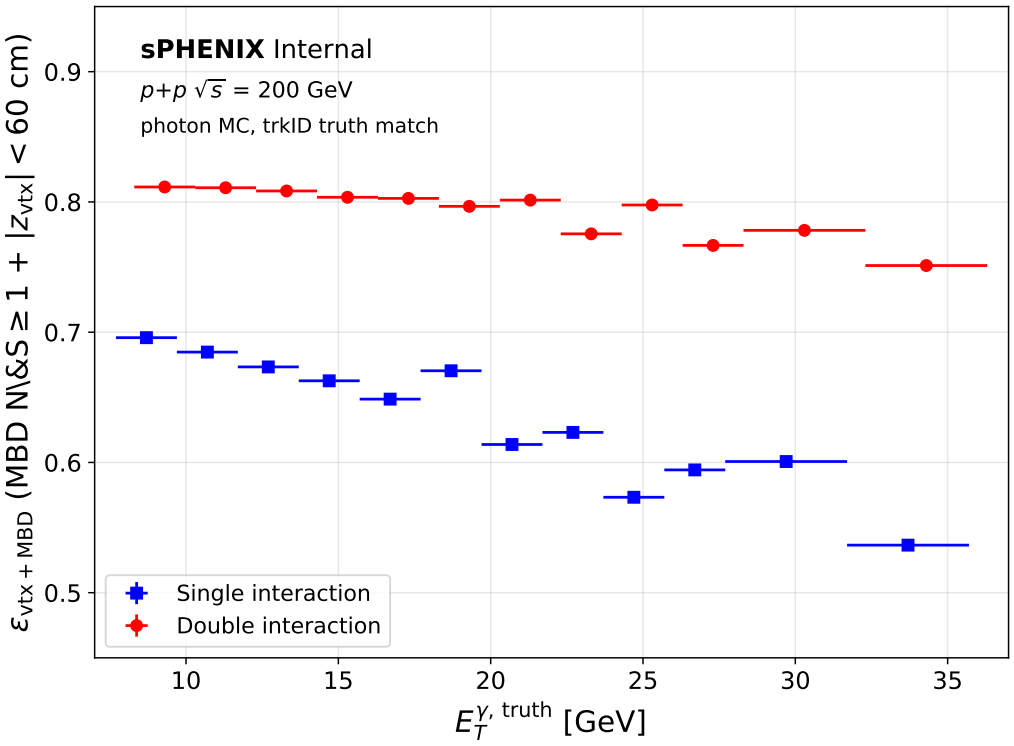

Figure 1: Per-p_{T} vertex+MBD acceptance for single-interaction (blue) and double-interaction (red) photon MC. Error bars are binomial. The double-interaction acceptance is 19% higher on average due to pileup MBD rescue and vertex centering.

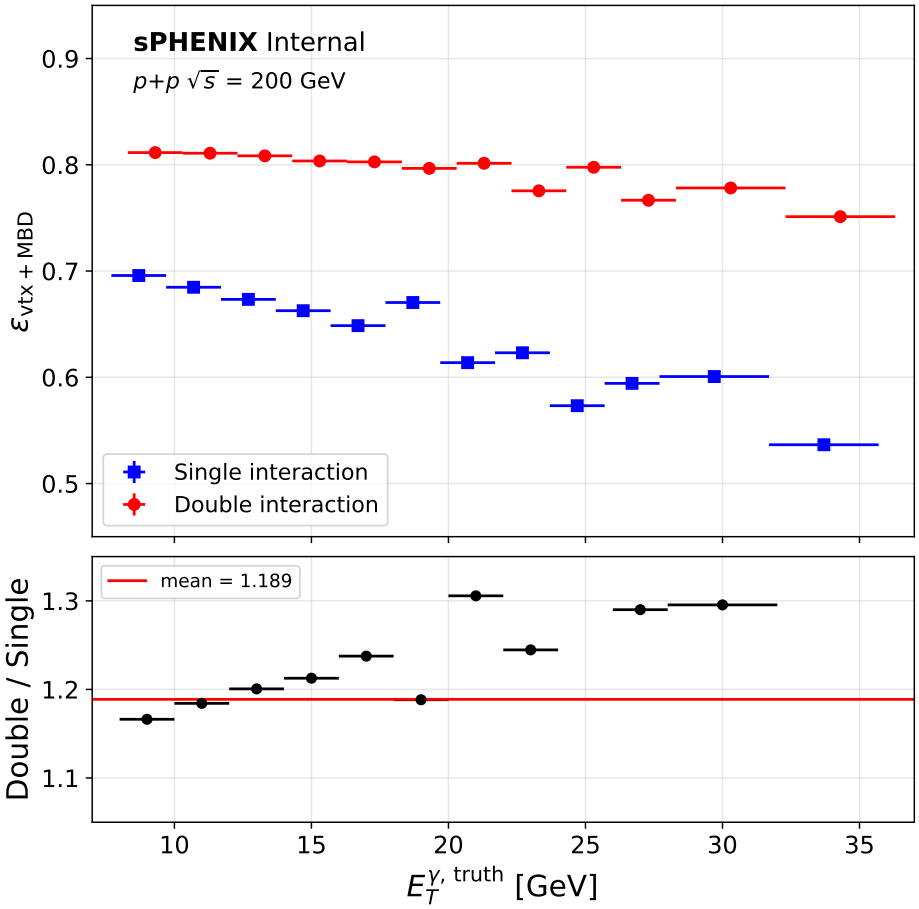

Figure 2: Top: vertex+MBD acceptance for both interaction types. Bottom: ratio (double/single), showing a ∼1.19 enhancement that increases slightly with p_{T} (from 1.17 at 8 GeV to 1.28 at 30 GeV).

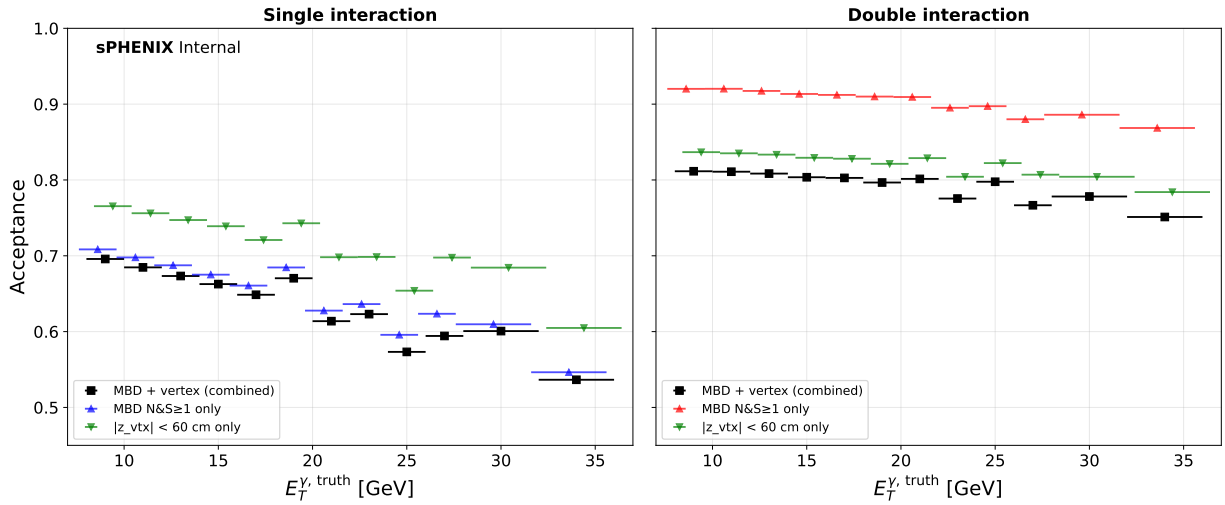

Figure 3: Acceptance breakdown into MBD-only (triangles up), vertex-only (triangles down), and combined (squares) for single (left) and double (right) interaction MC.

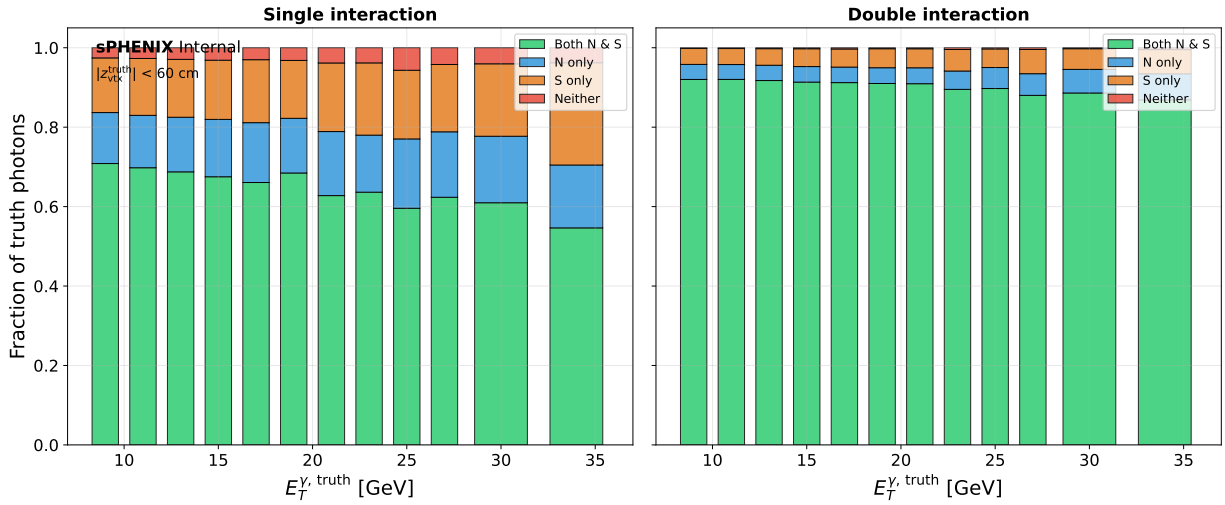

Figure 4: MBD hit decomposition for truth photons within the vertex acceptance. "Both N&S" = MBD coincidence satisfied; "N only" / "S only" = single-arm fire; "Neither" = no MBD hits. The double-interaction overlay nearly eliminates the "neither" category.

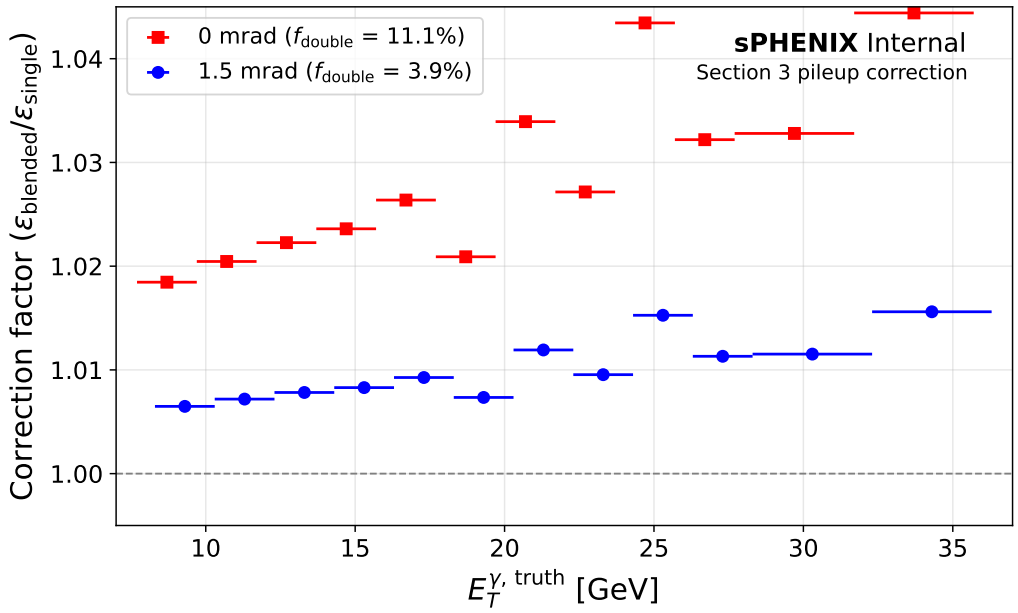

Figure 5: Per-p_{T} correction factor for 0 mrad (red, $f_{\text{double}} = 11.1\%$) and 1.5 mrad (blue, $f_{\text{double}} = 3.9\%$). The correction increases with p_{T} because the single-interaction acceptance drops faster at high p_{T} (forward photons more sensitive to vertex shifts), while double-interaction rescues proportionally more events.

4 Impact on the Cross-Section

The correction increases $\varepsilon_{\text{vtx+MBD}}$ in the efficiency denominator, which *decreases* the measured cross-section:

$$\left.\frac{d\sigma}{dp_{\mathrm{T}}}\right|_{\text{corrected}} = \left.\frac{d\sigma}{dp_{\mathrm{T}}}\right|_{\text{current}} \times \frac{\varepsilon^{\text{single}}_{\text{vtx+MBD}}}{\varepsilon^{\text{blended}}_{\text{vtx+MBD}}} \tag{3}$$

Table 1: Integrated correction factors and cross-section impact.

Period	f_{double}	**Integrated correction**	**Cross-section change**
0 mrad	11.1%	+2.1%	−2.1%
1.5 mrad	3.9%	+0.7%	−0.7%

The p_{T}-dependent correction ranges from +2.0% at 8 GeV to +3.7% at 26 GeV for the 0 mrad period.

5 Discussion

5.1 Comparison with other systematics

The Section 3 correction is smaller than the dominant PPG12 systematics:

- Purity systematic: 5–10%

- Energy scale: 3–5%
- σ_{MBD} uncertainty: $\sim$7% (flat)

For the 1.5 mrad nominal result, the +0.7% correction is negligible. For 0 mrad or combined, the +2.1% (rising to 3.5% at high p_{T}) is comparable to the energy scale systematic and should be applied or assigned.

5.2 Limitations

1. Only `photon10_double` exists as full GEANT double-interaction MC. The correction uses the same Pass_2 for all p_{T}, though the actual correction should be computed per truth p_{T} window (photon5/10/20).
2. Triple+ interactions ($n \geq 3$) are folded into the double fraction. At $\mu \approx 0.12$ (0 mrad), $\mathrm{Prob}_3 \approx 0.3\%$ — negligible.
3. The correction is computed as a single average over all runs, not per-segment. The μ variation within a fill introduces a $< 0.5\%$ residual uncertainty (Jensen's inequality).
4. The double-interaction vertex distribution is from GEANT4 simulation; any data-MC discrepancy in the beam vertex profile propagates.

6 Recommendation

1. **1.5 mrad (nominal):** No correction needed. Assign +0.7% as a one-sided systematic if desired.
2. **0 mrad or combined:** Apply the per-p_{T} correction from Table in Figure 5, or assign +2.1% (integrated) as a systematic. The p_{T}-dependent correction table is available in `results/pileup_vertex_acce`
3. **Implementation:** Multiply the current `vertex_eff` in `CalculatePhotonYield.C:1009` by the per-p_{T} correction factor for the appropriate crossing-angle period.

Files

File	Purpose
`efficiencytool/study_pileup_vertex_acceptance.py`	Analysis script
`plotting/plot_pileup_vertex_correction.py`	Plotting script
`results/pileup_vertex_acceptance.root`	ROOT output
`results/pileup_vertex_acceptance.csv`	Per-p_{T} table
`plotting/figures/pileup_vtx/*.pdf`	Figures